MIL-P-2880D

18 November 1976

SUPERSEDING
MIL-P-2880C
20 March 1969

MILITARY SPECIFICATION

PAPER, TEMPLATE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of template paper (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

MMM-A-260	-	Adhesive, Water-Resistant (For Sealing Waterproofed Paper).
PPP-B-1055	-	Barrier Material, Waterproofed, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9310

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STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI) PUBLICATIONS:

T402 os-70 - Standard Conditioning and Testing Atmospheres For Paper, Board, Pulp Handsheets and Related Products
T404 os-76 - Tensile Breaking Strength and Elongation of Paper and Paperboard (using Pendulum-Type Tester)
T411 os-76 - Thickness (Caliper) of Paper and Paperboard
T807 os-75 - Bursting Strength of Paperboard and Linerboard

(Copies of TAPPI Testing Methods may be obtained from the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Template paper shall consist of a base sheet, free from groundwood pulp, treated by such processes as to insure compliance with this specification.

* 3.2 Size and form. Template paper shall be furnished in rolls 78 ± 1/2 inches (198 ± 1.3 cm) or 105 ± 1/2 inches (267 ± 1.3 cm) in width as specified (see 6.2). Each roll shall contain a minimum of 2,000 square feet (185.8 m^2) in one continuous piece without splices. Unless otherwise specified, each roll shall be wound on a fiber core with an outside diameter of 4 ± 1/8 inches (10 ± 0.3 cm). The core length shall be not less than the width of the roll nor greater by more than 1/4 inch (0.6 cm). The rolls shall show no telescoping of more than 3/8 inch (0.95 cm).

3.3 Surface. Template paper shall have smooth surfaces with a dull glaze over a lighter colored underbody. Both surfaces shall be alike. When scribed with a template makers instrument, the surfaces shall exhibit fine lines which are clearly visible, distinct and cannot be eradicated when tested as specified in 4.3.2.

3.4 Color. The surfaces of the finished template paper shall approximate Color No. 22144 of FED-STD-595.

3.5 Flexibility. Template paper shall show no signs of splitting or cracking when tested as specified in 4.3.2.

3.6 Physical properties. The physical properties of template paper shall comply with the requirements of table I when tested as specified in 4.3.2.

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* TABLE I. Physical properties

Thickness, inches(mm), average 1/	0.019 (0.048 mm)
Tensile strength, pounds/inch width, (kN/m) minimum:	
Machine direction	100 (17.5)
Cross direction	55 (9.6)
Bursting strength, points, minimum	180
Dimensional stability, percent, maximum 2/	
Machine direction	0.30
Cross direction	1.00

1/ Thickness tolerance ± 0.0015 inch (0.038 mm)

2/ Percent change in length for a change in relative humidity from 27 ± 2 percent to 89 ± 2 percent at 73° ± 3.5°F (22.8 ± 1.9°C).

3.7 Workmanship. Template paper shall be clean, of uniform finish, and color. It shall be free from holes, cuts, tears, cracks, abrasions, creases, and folds. Rolls shall be evenly wound and edges shall not be uneven or crushed. Cores shall not be damaged.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) as specified in 4.3.1.5. The lot size, for the purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of rolls for the examination in 4.3.1.1, 4.3.1.2 and 4.3.1.3, and in units of shipping containers for 4.3.1.4.

4.3.1.1 Visual examination. The template paper shall be examined for defects listed below. The sample unit for this examination shall be one yard (0.91 m), full width of one roll. No more than one sample unit shall be taken from any one roll. Examine both sides.

Examine	Defects
Color	Does not approximate color number specified. Not uniform.
Surface (both sides)	Not smooth. Not uniform. Not dark, no dull glaze over a lighter colored underbody.
Workmanship	Not clean. Any hole, cut, tear, crack, abrasion, crease or fold. Any wrinkle, scuff or lump. Any ragged, uneven or crushed edge.

4.3.1.2 Examination of the end item for defects in dimensions. The sample unit for this examination shall be one roll.

Examine	Defect
Width of roll	Varies by more than ± 1/2 inch (1.29 cm) from specified width.

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Examine	Defect
Core length	Less than the specified roll width or greater by more than 1/4 inch (0.64 cm)
Core diameter	Outside diameter of core not within specified tolerances.

* 4.3.1.3 Examination of the end item for defects in construction. The sample unit for this examination shall be one roll.

Examine	Defects
Number of square feet per roll	Less than 2000 square feet (185.8 m^2) per single piece.
Winding of roll	Not wound evenly, tightly or smoothly causing wrinkles, creases or waves. Telescoping more than 3/8 inch (0.95 cm) on either roll edge.
Core	Core crushed, mutilated or collapsed.

* 4.3.1.4 Packaging inspection. An examination shall be made to determine that markings, materials and workmanship, comply with the requirements of section 5. The sample unit shall be one roll fully prepared for shipment.

Examine	Defect
Markings (exterior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any nonconforming component; component missing, damaged, or not as specified.
Workmanship	Not overwrapped as specified. Wrapper material other than specified for applicable packing. Headers, inside or out, missing; not of material specified. Closures not accomplished by specified or required methods or material.

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4.3.1.5 Inspection levels and acceptable quality levels (AQLs) for examination. The inspection levels, for determining the sample size and acceptable quality levels (AQLs), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.1.1	I	2.5
4.3.1.2	S-2	2.5
4.3.1.3	S-2	2.5
4.3.1.4	S-1	2.5

* 4.3.2 Testing of the end item. The end item shall be tested for the applicable characteristics in table II. The lot size shall be expressed in rolls of template paper. The sample unit shall consist of one square yard (0.836 m^2) of paper and no more than one sample unit shall be taken from any one roll. The sample unit shall be cut from the roll at least three feet (0.91 m) lineally from the outside or from the core and at least 3 inches (7.6 cm) from any edge. The sample unit shall be conditioned in accordance with the atmospheric conditions specified in TAPPI Method T402 0s-70 before cutting the test specimens to exact size, and all testing, except for flexibility, shall be performed under these conditions. All requirements shall be applicable to the sample unit. All test reports shall include the individual values used in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The lot size and sample size shall be as follows:

Lot size (rolls)	Sample size (No. of sample units)
800 or less	2
801 to 22,000 inclusive	3
22,001 and over	5

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INSTRUCTIONS FOR TESTING OF THE END ITEM

TABLE II

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per Unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver	Per sample unit	Pass or Fail	Numerically to Nearest		
Fiber identification	3.1	1/							
Thickness 2/	Table I	T411 os-68			10		0.0001 inch (0.0001 mm)		
Tensile strength									
Machine direction 2/	Table I	T404 os-74			10		pound (kg)		
Cross direction 2/	Table I	T404 os-74			10		pound (kg)		
Bursting strength 2/	Table I	T807 os-75			10 (5 per side)		point		
Dimensional stability									
Machine direction	Table I	4.4.3			2		0.01 percent		
Cross direction	Table I	4.4.3			2		0.01 percent		
Flexibility									
Machine direction	3.5	4.4.1			1	X			
Cross direction	3.5	4.4.2			1	X			
Surface	3.3	4.4.2			1	X			

1/ A certificate of compliance will be acceptable for the requirement stated.

2/ For this characteristic the number in the test method column refers to a TAPPI test number.

AMXRE Form 598
8 February 1965

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4.4 Test procedures.

* 4.4.1 Flexibility. Cut two specimens, 3 inches (7.6 cm) wide by 12 inches (30.5 cm) long, from the sample unit. One of the specimens shall be cut with the long dimension parallel to the machine direction of the template paper and the other, with the long dimension parallel to cross machine direction. Condition both specimens for 7 hours at 32° ± 2°F (0 ± 1°C). Immediately after the conditioning period, bend the specimens around a 3 inch (7.6 cm) diameter mandrel and observe for cracking or splitting.

* 4.4.2 Surface. Cut a specimen 2 inches (5 cm) wide by 12 inches (30.5 cm) long from the sample unit. Using a template makers scribing instrument, draw five consecutive lines, one-sixteenth of an inch (0.16 cm) apart and 10 inches (25.4 cm) long on each side of the specimen. The result shall be considered unsatisfactory if the lines are not lighter in color than the surface, clearly visible and distinct. Rub over the lines vigorously with the fingers or a cloth: eradication of the lines shall be cause for rejection.

4.4.3 Dimensional stability.

* 4.4.3.1 Apparatus. The apparatus shall consist of (1) a cabinet in which changes in the length of one or more specimens of paper are produced by controlled variations of relative humidity and measured by means of an extensometer (or other suitable means of equivalent accuracy) capable of measuring to within 0.0001 inch (0.0025 mm), (2) a means for holding the specimen in the test chamber such as a set of jaws, clamps or an upper clamp and a T-hook, having a gripping surface of 1/2 inch (1.3 cm) and (3) means of subjecting the specimen to a tension stress of 7 ounces (198 g).

* 4.4.3.2 Test specimens. Duplicate specimens shall be used for each test. One set of specimens each measuring 1 inch (2.5 cm) wide by 10-12 inches (25-30 cm) long, shall be cut with the long dimension parallel to the machine direction. Another set shall be cut as before, but with the long dimension parallel to the cross direction. The specimens shall be cut with the edges clean and parallel.

* 4.4.3.3 Procedure. Attach the specimen to the holding device so that the distance between the jaws is 10 inches (25 cm) and the 1/2 inch (1.25 cm) gripping edge is 1/4 inch (0.64 cm) from each edge. Apply a tension of 7 ounces (198 g) to the specimen. Condition the specimen at a relative humidity of 89 ± 2 percent for 6 hours. The temperature of the cabinet shall be maintained at 73° ± 3.5°F (22.8 ± 1.9°C) throughout the test period. After

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the specimen has been conditioned as above, adjust the test cabinet to 27 ± 2 percent relative humidity and hold the specimen under these conditions for 48 hours. After this time interval, adjust the extensometer (or measuring device) to give a zero reading. Adjust the cabinet to 89 ± 2 percent relative humidity as before and hold the specimen under these conditions for 48 hours. Record the extensometer reading. Using the average change in length for duplicate specimens, calculate the dimensional stability to two decimal places as the percentage change in length based on an initial length of 10 inches (25.4 cm) from 27 ± 2 to 89 ± 2 percent relative humidity.

5. PACKAGING

5.1 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.1.1 Level A packing. Each roll of template paper shall be completely wrapped in a minimum of two thicknesses of barrier material conforming to PPP-B-1055. Treated surfaces of the barrier material shall not come in contact with the template paper. The wrapper shall be closed at the ends by means of inside and outside headers made of the same material as the wrapper. The seams of the wrapper and the outer headers shall be sealed with an adhesive conforming to MMM-A-260 in a manner which will effect a watertight seal and result in no bleeding of asphalt in the waterproofed paper.

5.1.2 Level B packing. Each roll of template paper shall be wrapped overall with a minimum of two thicknesses of kraft paper, 60 pounds (27.2 kg) minimum basis weight (24 x 36 - 500). The wrap shall be closed at the ends by means of inside and outside headers. The seams of the wrap and the outer headers shall be securely sealed with an adhesive commercially used for this purpose.

5.1.3 Commercial packing. Rolls of template paper shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Container shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.2 Marking. In addition to any special marking required by the contract, wrapped rolls shall be marked in accordance with MIL-STD-129. Marking for Commercial shipments shall be the same as that specified for level C.

6. NOTES

6.1 Intended use. Template paper covered by this specification is intended for use in making templates or patterns for ships' plates.

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* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Width of rolls.
(c) Core diameter (if other than 4 inches (10 cm)) (see 3.2).
(d) Selection of applicable levels of packing (see 5.1).

* 6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, corrections, modifications, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 99

User activities:

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Preparing activity:

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